Ultimate DSA Placement Dashboard - SRS Version 3

Volume 3: Curated High-ROI Problem Database Blueprint

Purpose

Define the structure for a curated database of approximately 300–350 placement-focused DSA problems. The final implementation should avoid duplicates by mapping equivalent Striver, LeetCode and GFG problems into one master entry.

Category	Target Count	Priority
Arrays	35	★★★★★
Hashing	20	★★★★★
Strings	20	★★★★★
Binary Search	25	★★★★★
Sliding Window	18	★★★★★
Two Pointer	15	★★★★★
Linked List	20	★★★★☆
Stack & Queue	25	★★★★☆
Greedy	20	★★★★☆
Binary Trees	25	★★★★☆
BST	12	★★★☆☆
Heap	12	★★★☆☆
Graphs	20	★★★☆☆
Dynamic Programming	20	★★★☆☆
Bit Manipulation	10	★★★☆☆
Math/Misc	10	★★☆☆☆

Required Fields Per Problem

Problem ID, Problem Name, Category, Pattern, Difficulty, ROI (1–10), Frequency (★), Company Tags,
Striver Link, LeetCode Link, GFG Link, YouTube Link, Time Taken, Hint Used, Confidence, Revision Count,
Next Revision Date, Notes.

Pattern Taxonomy

Each problem must belong to one or more patterns:
Hash Map, Prefix Sum, Sliding Window, Two Pointer, Binary Search, Binary Search on Answer,
Kadane, Monotonic Stack, Greedy Interval, DFS, BFS, Tree Traversal, LCA, Heap,

Backtracking,
Union Find, Topological Sort, Knapsack, LIS, Matrix DP.

ROI Rules

ROI 10: Mandatory for almost every placement.
ROI 8–9: High frequency.
ROI 6–7: Medium frequency.
ROI 4–5: Good enrichment.
ROI 1–3: Low priority; postpone until after placements.

Company Mapping

Every problem should optionally carry tags such as:
Accenture, Deloitte, Infosys DSE, Cognizant Next, Capgemini, EY, Oracle, JPMC.
This enables company-specific revision filters.

Filtering Requirements

Support filters by Topic, Pattern, Difficulty, ROI, Frequency, Status, Confidence, Company Tag and Next Revision Date.

Version 4 Preview

Next document will define dashboard wireframes, Excel formulas,
conditional formatting, charts, analytics and automation logic.